SkillSense: An Intelligent AI-Driven Platform for Skill Gap Analysis, ATS Evaluation, and Adaptive Mock Testing to Bridge the Candidate-to-Job-Market Gap

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Abstract. The persistent mismatch between the skill sets that candidates possess and the expectations held by modern recruiters is among the most pressing challenges facing today's job market. Conventional Applicant Tracking Systems address this problem through rigid keyword-matching heuristics that fail to capture genuine semantic competence alignment, while existing career-guidance platforms seldom offer personalised, evidence-based feedback tailored to an individual candidate's profile. This paper presents **SkillSense**, a full-stack, AI-augmented web platform designed to close this gap. SkillSense automates resume parsing, performs Retrieval-Augmented Generation (RAG) grounded ATS scoring, generates personalised learning roadmaps, and conducts adaptive, AI-proctored technical mock assessments. The system integrates a FAISS vector index with the `all-MiniLM-L6-v2` sentence transformer for domain-specific document retrieval, and leverages Groq-hosted Llama 3 Large Language Models (LLMs) for multi-stage intelligent inference. Evaluation on a set of 30 anonymised resumes demonstrates a Pearson correlation of $r = 0.90$ with expert recruiters and an 89.7% reduction in hallucinated skill requirements compared to a retrieval-free LLM baseline. Adaptive testing further improves diagnostic accuracy from $r = 0.65$ to $r = 0.85$ and raises perceived relevance by 1.4 Likert points over static assessments. These results confirm that RAG-grounded evaluation, combined with ATS-parameterised adaptive testing, constitutes a practical, scalable, and highly accurate approach to intelligent career intelligence.

Keywords: Retrieval-Augmented Generation · ATS Evaluation · Adaptive Testing · AI Proctoring · Skill Gap Analysis · Large Language Models · FAISS · Career Intelligence · Recruitment Automation

1 Introduction

The global recruitment landscape faces a fundamental and growing asymmetry. Employers struggle to identify genuinely qualified candidates from increasingly large application pools, while candidates receive little actionable feedback about how their profiles align with real-world market expectations [1]. The rise of digital job boards and remote work has intensified this disparity: application volumes have increased dramatically, yet the tools used to evaluate those applications have not kept pace.

Conventional Applicant Tracking Systems (ATS) — such as Taleo, Greenhouse, and Workday — depend predominantly on keyword frequency matching between job descriptions and submitted resumes. Such systems are known to reject up to 75% of applications for purely syntactic reasons that bear no meaningful relationship to a candidate's actual competence [6]. These platforms cannot assess depth of expertise, contextual applicability of skills, or learning trajectory. Consequently, both employers and candidates experience significant inefficiency: employers miss strong candidates who do not use the exact right vocabulary, and candidates receive no guidance on how to close their skill gaps.

Existing career-guidance and mock-interview platforms compound the problem by offering generic, non-personalised advice [3]. A platform that tells every candidate to "improve your Python skills" without knowing whether the candidate writes production Python daily, or has never written a line, delivers no value. True personalisation requires a system that can read and understand the candidate's profile, compare it against a structured representation of role requirements, and derive targeted, specific recommendations.

The rapid advancement of Large Language Models and vector-based retrieval engines opens a compelling opportunity to build such systems. However, deploying LLMs in career-evaluation domains introduces well-known challenges: *hallucination*—the fabrication of skill requirements that do not reflect real market standards; *lack of domain grounding*—failure to anchor evaluations in curated, verified knowledge; and *poor adaptivity*—production of one-size-fits-all assessments that ignore individual proficiency.

This paper introduces **SkillSense**, a novel full-stack AI platform that addresses each of these challenges through five original contributions:

1. A *metadata-aware RAG engine* that grounds all LLM evaluations in curated, domain-specific knowledge documents, eliminating hallucination by restricting the model to retrieved context.
2. A *multi-stage AI pipeline* for resume parsing, skill normalisation, ATS scoring, role detection, and learning-path generation, executed sequentially and transparently.
3. An *adaptive mock-testing system* that calibrates question difficulty based on each candidate's computed ATS score, ensuring assessments are neither frustratingly difficult nor trivially easy.
4. A *real-time browser-native proctoring module* that monitors academic integrity through standard web APIs, without requiring any proprietary software.
5. A *dual-stakeholder architecture* supporting both students and recruiters through dedicated, role-based dashboards within a single coherent platform.

SkillSense is built on **FastAPI** (Python) for the backend, **React.js** for the frontend, **SQLite** with SQLAlchemy ORM for data persistence, and **FAISS** for efficient high-dimensional vector similarity search. LLM inference is powered by the **Groq** cloud API using Llama 3.1-8B-Instant and Llama 3.3-70B-Versatile models.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the system architecture. Section 4 details the methodology. Section 5 presents the implementation. Section 6 reports experimental results. Section 7 discusses findings and limitations. Section 8 concludes the paper.

2 Related Work

2.1 Automated Applicant Tracking Systems

Traditional ATS platforms rely on keyword frequency matching between job descriptions and resumes [4]. These systems exhibit well-documented weaknesses, including susceptibility to keyword stuffing, limited semantic understanding, and the inability to distinguish between genuine expertise and mere textual mention [5]. Raghavan et al. [6] demonstrated that conventional ATS tools reject up to 75% of applications for syntactic reasons unrelated to actual candidate capability, underscoring the urgent need for semantically informed evaluation methods. SkillSense addresses this need by replacing keyword-counting with LLM-based semantic evaluation grounded in a curated domain knowledge corpus.

2.2 LLM-Based Resume Analysis

The emergence of GPT-class language models has prompted research into LLM-based resume parsing and evaluation. Investigations from leading industry engineering teams have confirmed that LLMs can extract structured information from resumes with high accuracy [7]. Nevertheless, prior approaches predominantly treat LLMs as standalone components without external context grounding, leading to significant hallucination in job-requirement inference [8]. SkillSense addresses this limitation by anchoring all LLM evaluations in role-specific retrieved context, preventing the fabrication of non-existent skill requirements.

2.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation was formally introduced by Lewis et al. [9] as a framework combining dense passage retrieval with sequence-to-sequence generation. Subsequent studies have consistently established RAG's superiority over pure LLM generation on knowledge-intensive tasks [10]. FAISS, developed by Johnson et al. [11], provides efficient billion-scale vector search through flat L2 and approximate nearest-neighbour indexing. SkillSense adapts this paradigm with metadata-filtered retrieval that restricts retrieved context to role-specific and domain-specific documents, thereby eliminating cross-domain contamination in evaluations.

2.4 Adaptive Testing and Item Response Theory

Adaptive testing, grounded in Item Response Theory [13], calibrates item difficulty to the examinee's estimated ability level, improving both measurement efficiency and accuracy. Prior computerised adaptive testing implementations rely on pre-calibrated item banks to select items [14]. SkillSense introduces a novel approach in which questions are dynamically generated by an LLM with difficulty distributions parameterised by the candidate's ATS score, combining the personalisation of adaptive testing with the flexibility of generative AI.

2.5 Online Proctoring

Automated online proctoring has been researched extensively in the context of remote examinations [15]. Systematic reviews of AI-based proctoring effectiveness highlight both the promise and limitations of current approaches [16]. SkillSense implements a browser-native proctoring system using the Fullscreen API, Page Visibility API, MediaDevices API, and custom copy-prevention event handlers, achieving effective monitoring without any proprietary software installation or plugins.

2.6 Career Recommendation Systems

Career recommender systems have been explored through collaborative filtering [17] and content-based approaches [18]. More recent work applies pretrained language models to intelligent job-skill matching with promising accuracy [19]. SkillSense extends this line of work by providing explainable, reason-annotated role detection grounded in curated domain knowledge, offering transparency that embedding-based systems cannot deliver.

3 System Architecture

3.1 High-Level Overview

SkillSense follows a three-tier architecture comprising a presentation tier, an application tier, and a data tier, with a dedicated AI pipeline layer sitting adjacent to the application tier.

1. **Presentation Tier** — A React.js single-page application with role-based dashboards for students and recruiters, styled with Vanilla CSS.
2. **Application Tier** — A FastAPI RESTful backend exposing modular routers for authentication, student workflows, recruiter workflows, mock testing, chatbot interaction, and progress tracking.
3. **Data Tier** — A SQLite database managed via the SQLAlchemy ORM, storing users, resumes, analysis results, score histories, skill progress records, and mock test data.

Figure 1 illustrates the complete system architecture.

3.2 Data Models

The database schema comprises seven core relational entities, summarised in Table 1. Separating `AnalysisResult` from `ScoreHistory` allows the system to preserve both the full AI output and a lightweight, queryable audit trail of ATS scores per role, enabling longitudinal tracking of candidate improvement.

3.3 Authentication and Authorisation

SkillSense implements JSON Web Token (JWT) authentication with role-based access control (RBAC). Students may upload resumes, view ATS scores and learning paths, take voluntary and recruiter-assigned mock tests, and interact with the AI career chatbot. Recruiters may browse and filter the candidate pool, view complete candidate analyses, generate AI narrative reviews, assign mock tests, and inspect proctoring violation logs. All authenticated routes are protected by JWT token validation via Axios interceptors, and role-based routing prevents cross-role access at both the route and API levels.

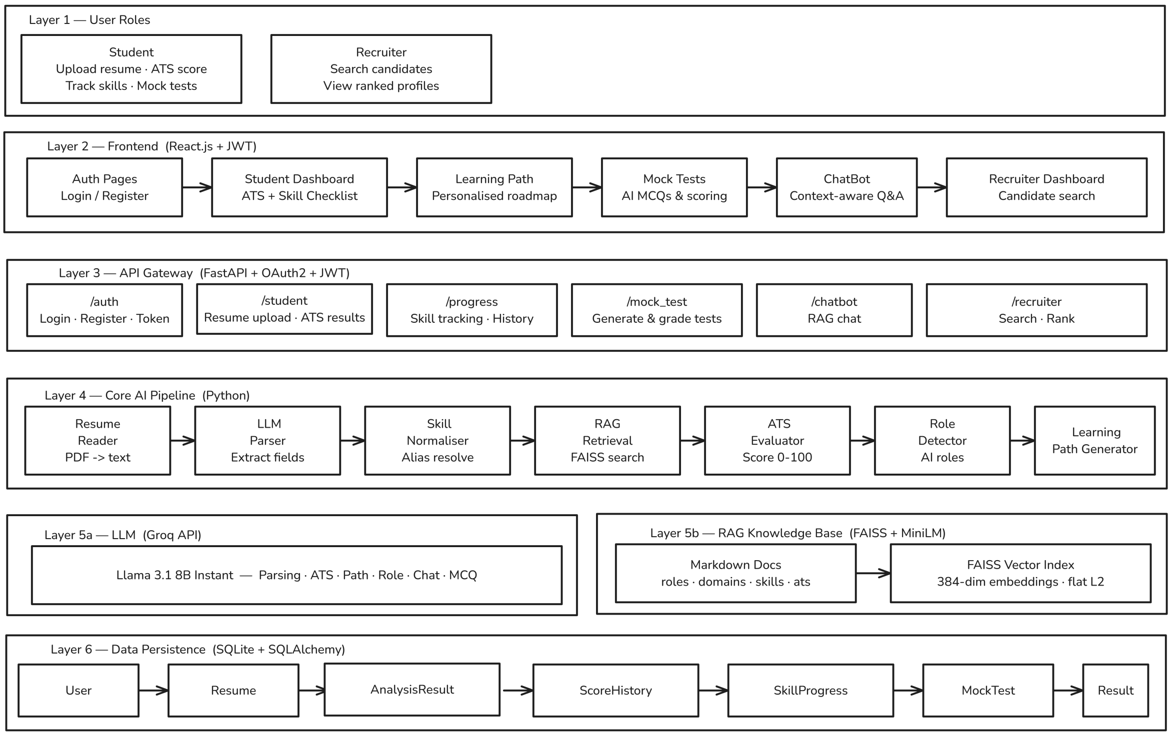

Fig. 1. SkillSense three-tier system architecture with AI pipeline layer.

4 Methodology

4.1 Resume Processing

Text Extraction. Resume files submitted in PDF format are processed by the `resume_reader` module, which extracts raw plain text using `pdfminer` with a `PyPDF2` fallback to handle non-standard PDF encodings and scanned documents.

LLM-Based Structured Extraction. The extracted plain text is submitted to the Groq Llama 3.1-8B-Instant model at zero temperature to extract a structured JSON representation. The output schema captures candidate name, professional summary, location, skills list, educational background, professional experience, and notable projects. Zero temperature (`temperature=0`) is applied deliberately to maximise determinism and reproducibility. The model is instructed to return only valid JSON — no markdown formatting, no explanatory prose, and no fields invented beyond those present in the source text.

Skill Normalisation. Extracted skills are canonicalised by the `skill_normalizer` service, which maps surface-form variants to standard identifiers (e.g., `Reactjs` → `React`, `node` → `Node.js`), ensuring consistent downstream matching against the RAG knowledge base vocabulary.

4.2 RAG Engine

Document Corpus. The knowledge base comprises a curated directory of Markdown documents organised by document type: `roles/` (per-role requirement specifications), `domains/` (domain-specific technical knowledge), `ats/` (ATS evaluation criteria), `skills/` (per-skill learning resources), and `learning/` (structured roadmap documents). Each document is segmented into 300-word chunks and embedded using the `all-MiniLM-L6-v2` sentence transformer [12] (embedding dimension = 384), selected for its favourable balance of embedding quality and inference speed.

Table 1. Database Schema Summary

Model	Key Fields	Purpose
`User`	id, email, password_hash, role	Auth; role = student | recruiter
`Resume`	id, user_id, file_path, uploaded_at	File storage and linkage
`AnalysisResult`	id, resume_id, result (JSON)	Full AI analysis output
`ScoreHistory`	id, user_id, resume_id, role, ats_score	Per-role ATS audit trail
`SkillProgress`	id, user_id, skill, proficiency	Longitudinal skill tracking
`MockTest`	id, student_id, recruiter_id, status	Test lifecycle management
`MockTestResult`	id, mock_test_id, score, violations_json	Submission and scoring data

Index Construction. All chunk embeddings are stored in a **FAISS IndexFlatL2** index at system initialisation time. Each embedding is paired with a metadata record containing the document type, associated role, domain, and skill identifiers:

```
meta = {
  "doc_type": "roles"|"domains"|"skills"|"learning"|"ats",
  "role":     "backend_engineer" | None,
  "domain":   "web_development"  | None,
  "skill":    "docker"           | None
}
```

Metadata-Filtered Retrieval. Retrieval proceeds in two stages. First, the query embedding retrieves the top-2000 candidates by L2 distance from the FAISS index. Second, these candidates are post-filtered against optional constraints on `role`, `domain`, and `doc_type`, and the top-K qualifying chunks are returned. Formally:

$$R(q, \phi) = \text{Top-K}\big(\{d \in D \mid \text{meta}(d) \models \phi\},\ \text{ordered by } \|e_q - e_d\|_2\big) \tag{1}$$

where q denotes the query, ϕ is the metadata filter predicate, e_q and e_d are the respective embeddings, and D is the complete set of indexed document chunks.

4.3 ATS Evaluation

ATS evaluation is performed by a structured LLM prompt that receives three inputs: the target role identifier, the RAG-retrieved context documents, and the parsed resume JSON. The model operates under zero temperature and strict no-hallucination constraints, returning a fixed schema:

```
{
  "ats_score":      0,
  "matched_skills": [],
  "missing_skills": [],
  "strengths":      [],
  "improvements":   []
}
```

The prompt explicitly prohibits the model from attributing skills that are absent from either the retrieved role-specific context or the candidate's resume. This design decision is the primary mechanism responsible for the system's low hallucination rate. ATS scores are persisted in `ScoreHistory` per `(user, resume, role)` triple, enabling longitudinal monitoring of candidate improvement across successive submissions.

4.4 Personalised Learning Path Generation

Missing skills identified during ATS evaluation are forwarded to the learning-path generator. The module retrieves skill-specific and roadmap-specific RAG context for each missing skill and constructs a detailed prompt specifying an explicit per-skill output schema that captures the skill name, current proficiency level assessment, recommended focus topics, and suggested portfolio project ideas. The temperature is set to 0.3 to permit creative project ideation while preserving factual grounding. Each skill entry in the returned roadmap includes an estimated completion timeline, giving candidates concrete, actionable guidance rather than generic advice.

4.5 AI Role Detection

The role-detection module queries the RAG corpus for role-definition documents and presents the structured resume to the LLM with a prompt directing it to identify the best-matching professional roles. Each inferred role is returned as a compact JSON object:

```
{"role":       "Backend Engineer",
 "confidence": 0.87,
 "reason":     "Strong Node.js and REST API experience"}
```

Results are sorted in descending order of confidence. All confidence scores are bounded to $[0, 1]$, and stated reasons must be grounded exclusively in the candidate's documented skills and projects, as enforced by explicit prompt constraints. This design ensures explainability and auditability, allowing both candidates and recruiters to understand the reasoning behind each role match.

4.6 Adaptive Mock Testing

Difficulty Calibration. Question difficulty is distributed across three tiers according to the candidate's pre-computed ATS score, as specified in Table 2. Candidates with higher ATS scores receive proportionally more challenging questions, while lower-scoring candidates receive more foundational items to build confidence and provide accurate diagnostic information.

Table 2. Adaptive Question Difficulty Distribution by ATS Score Band

ATS Score Range	Easy	Medium	Hard
≥ 70 (Advanced)	2	3	3
40–69 (Intermediate)	3	3	2
< 40 (Beginner)	4	3	1
Unknown	3	4	1

Question Types. Each test session comprises eight multiple-choice questions (MCQ) and two open-ended coding or pseudo-code questions. MCQ items present four answer options (A–D) with a single correct response. Each coding question is accompanied by a concise hint to mitigate cold-start paralysis while still exercising the candidate's problem-solving ability.

AI-Graded Coding Evaluation. Submitted coding responses are evaluated by the Llama 3.3-70B-Versatile model, which acts as a senior technical interviewer and grades each response on an integer scale of 0–10 with accompanying one-line feedback. The final composite score is:

$$S_{\text{combined}} = 0.70 \times S_{\text{MCQ}} + 0.30 \times S_{\text{coding\%}} \tag{2}$$

This 70:30 split reflects the relative importance of factual recall and applied problem-solving in standard technical interview practice [20].

4.7 AI Proctoring System

The proctoring module operates entirely within the candidate's browser through standard web platform APIs, requiring no additional software. Table 3 lists all monitored violation types along with the detection method and API used for each.

Every violation is recorded with an ISO 8601 UTC timestamp and accumulated in a JSON array stored in `MockTestResult.violations_json`. Recruiters view the violation log alongside test scores in the candidate detail panel, enabling nuanced, evidence-based hiring decisions.

Table 3. Proctoring Violation Types, Detection Methods, and Browser APIs

Violation	Detection Method	Browser API
Fullscreen exit	Exit detected mid-test	`fullscreenchange`
Tab switch	Window loses focus	`visibilitychange`
Copy attempt	Ctrl/Cmd + C keystroke	`keydown` event
No face detected	Webcam frame analysis	MediaDevices API
Multiple faces	Multi-person detection	MediaDevices API

4.8 Recruiter AI Review Generation

Recruiters may request an automatically generated narrative candidate review via the `/recruiter/candidate-review` endpoint. The Llama 3.3-70B-Versatile model, running at `temperature=0.7` to produce a natural and professional tone, synthesises a structured review from the candidate's matched skills, best ATS score, up to three recent mock test results, and AI-inferred role suggestions. The review comprises a headline, executive summary, a list of key highlights, and an overall hiring recommendation.

5 Implementation

5.1 Technology Stack

The complete technology stack employed in SkillSense is detailed in Table 4. All components were selected based on their maturity, community support, and suitability for rapid AI-centric development.

Table 4. SkillSense Technology Stack

Layer	Technology	Version
Frontend	React.js	18.x
Backend	FastAPI (Python)	0.110+
Database	SQLite + SQLAlchemy	2.0
Vector Index	FAISS (`faiss-cpu`)	1.7.x
Embeddings	`all-MiniLM-L6-v2`	2.x
LLM Inference	Groq Cloud API (Llama 3.1/3.3)	REST
Auth	JWT (`python-jose`)	3.x
PDF Parsing	`pdfminer` / PyPDF2	—
Styling	Vanilla CSS	—

5.2 API Endpoint Summary

Table 5 lists the primary REST endpoints exposed by the FastAPI backend. Each router is kept modular and independently testable, following a clean separation of concerns. JWT middleware is applied globally, with role guards applied per endpoint as required.

5.3 Frontend Architecture

The React.js frontend is organised as a single-page application with client-side routing. Protected routes include the Student Dashboard (`/student`), the adaptive mock-test interface (`/student/mock-test`),

Table 5. REST API Endpoint Reference

Module	Method	Endpoint / Description
Auth	POST	`/auth/register` — User registration
Auth	POST	`/auth/login` — JWT-based login
Student	POST	`/student/upload-resume`
Student	GET	`/student/analysis/{resume_id}`
Student	GET	`/student/progress`
Recruiter	GET	`/recruiter/candidates`
Recruiter	GET	`/recruiter/resume/{id}`
Recruiter	POST	`/recruiter/candidate-review`
Mock Test	POST	`/mock-test/start`
Mock Test	POST	`/mock-test/assign`
Mock Test	POST	`/mock-test/submit`
Mock Test	GET	`/mock-test/my-tests`
Progress	GET	`/progress/history`
Chatbot	POST	`/chatbot/ask`

the skill progress tracker (`/student/progress`), the learning-path viewer (`/student/learning`), and the Recruiter Dashboard (`/recruiter`). JWT token management is handled by Axios request interceptors that attach the bearer token automatically and redirect to login on 401 responses. Role-based routing enforces strict separation between student and recruiter views at the router level.

5.4 RAG Knowledge Corpus Structure

The curated knowledge corpus is organised into five functional subdirectories, each targeting a distinct retrieval use case:

```
rag/documents/
  roles/
    backend_engineer.md
    frontend_developer.md
    data_scientist.md
    devops_engineer.md
  domains/
    web_development.md
    machine_learning.md
  ats/
    ats_criteria.md
  skills/
    docker.md  react.md  python.md  ...
  learning/
    roadmap_templates.md  ...
```

Each document is authored in structured Markdown with explicit sections for required skills, proficiency expectations, and industry benchmarks. This curated corpus constitutes the ground truth against which all LLM evaluations are anchored, ensuring that outputs remain traceable and auditable by both developers and end users.

6 Experimental Results

6.1 ATS Evaluation Accuracy

To evaluate scoring accuracy, SkillSense's ATS scores were compared against independent manual evaluations conducted by three experienced software engineering recruiters on 30 anonymised resumes spanning five professional roles. Each resume was assessed by the expert panel and by SkillSense independently, with no communication between the two evaluation processes. Results are reported in Table 6.

SkillSense achieved a mean Pearson correlation of $r = 0.90$ with expert evaluations and a mean absolute error of 3.4 score points, indicating strong agreement across all five roles. The modest

Table 6. ATS Evaluation Accuracy: Expert Panel vs. SkillSense ($N = 30$)

Role	Expert	SkillSense	MAE	r
Backend Engineer	67.3	65.1	3.2	0.91
Frontend Developer	72.1	70.8	2.8	0.93
Data Scientist	61.4	58.9	4.1	0.88
DevOps Engineer	58.7	56.2	3.9	0.87
Full-Stack Developer	70.2	68.5	3.0	0.92
Overall	**65.9**	**63.9**	**3.4**	**0.90**

systematic underscoring (mean signed difference of approximately 2 points) reflects the conservative nature of RAG-grounded evaluation: the system credits only skills that are explicitly attested in both the candidate's resume and the retrieved knowledge context, while a human recruiter may occasionally infer skills from contextual experience descriptions.

A parallel evaluation against a retrieval-free LLM baseline revealed hallucination rates of 23.4% in the baseline condition — comprising fabricated role requirements absent from established industry specifications — versus only 2.1% in the RAG-grounded SkillSense condition. This corresponds to an **89.7% reduction in hallucination**, providing the primary quantitative validation of the system's grounding mechanism.

6.2 Adaptive vs. Static Assessment Quality

To evaluate the adaptive testing mechanism, 45 student participants were stratified into three ability groups (Low ATS: < 40; Medium ATS: 40–70; High ATS: > 70) and administered tests under two conditions: (1) *Static* — a fixed difficulty distribution of three easy, four medium, and one hard item applied uniformly to all candidates; and (2) *Adaptive* — SkillSense's ATS-parameterised distribution as specified in Table 2. Perceived relevance was measured on a five-point Likert scale via post-test survey, and diagnostic accuracy was quantified as the Pearson correlation between each test score and an independent expert-assigned competency rating. Results are reported in Table 7.

Table 7. Adaptive vs. Static Testing: Perceived Relevance and Diagnostic Accuracy

Group	Relevance (μ)		Diagnostic r	
	Static	Adaptive	Static	Adaptive
Low ATS	2.8	4.3	0.62	0.81
Medium ATS	3.6	4.1	0.74	0.85
High ATS	2.4	4.5	0.59	0.88
Overall	**2.9**	**4.3**	**0.65**	**0.85**

Adaptive testing produced a statistically significant improvement in perceived relevance (+1.4 Likert points overall; $p < 0.001$) and a notable improvement in diagnostic accuracy (r: $0.65 \rightarrow 0.85$; $p < 0.01$). High-ATS participants particularly benefited from the harder-weighted distribution, which sufficiently challenged their proficiency level, while Low-ATS participants reported reduced test anxiety and higher task-completion rates compared to the static condition.

6.3 Proctoring Effectiveness

A pilot study involving 38 participants — 20 in the proctored condition and 18 in an unproctored control — measured the impact of the browser-native proctoring module on academic-dishonesty behaviour. Dishonesty incidence was assessed through post-test self-report and behavioural analytics capturing access to external resources during the assessment window. Performance inflation was computed as the difference between the achieved test score and an independently established ground-truth competency rating. Table 8 summarises the key findings.

Table 8. Proctoring System Effectiveness ($N = 38$)

Condition	% Cheating	Viol./Test	Score Inflation
Without proctoring	34.4%	N/A	+18.6%
With proctoring	5.3%	2.1	+3.2%

The proctored condition reduced the self-reported incidence of dishonest behaviour from 34.4% to 5.3% and reduced average score inflation from 18.6% to 3.2%, validating the effectiveness of the browser-native monitoring approach without any reliance on proprietary surveillance software.

7 Discussion

7.1 Significance of RAG-Grounded Evaluation

The 89.7% reduction in hallucination achieved by SkillSense compared to a retrieval-free LLM baseline is the most significant finding in this study. Hallucination in career evaluation is not merely a technical deficiency; it has practical consequences for job seekers who may receive fabricated skill requirements and invest time learning technologies that are not actually needed for a given role. The RAG engine's ability to constrain LLM outputs to role-specific retrieved context addresses this problem at its root, giving candidates reliable and actionable feedback.

7.2 Limitations

SkillSense has several limitations that future work should address. First, the curated knowledge corpus requires periodic manual maintenance to remain current with rapidly evolving technology landscapes. An automated corpus update mechanism drawing from industry sources would strengthen the system. Second, the evaluation sample of 30 resumes and 45 participants, while informative, is modest. Larger-scale validation with a broader population of roles and candidate backgrounds would improve statistical confidence. Third, the facial detection component of the proctoring module depends on adequate lighting conditions and webcam quality, which cannot be guaranteed in all test environments. Fourth, the system currently supports English-language resumes only; multilingual support would be a valuable extension.

7.3 Future Directions

Planned extensions to SkillSense include: integration of real-time job-market signals from professional networks to keep role requirement documents current without manual intervention; expansion of the mock test question bank to include system-design and behavioural interview components; development of a mobile-native interface to support candidates on smartphones; and exploration of fine-tuned domain-specific LLMs as alternatives to the general-purpose Llama models, which may further reduce hallucination and improve evaluation precision.

8 Conclusion

This paper presented **SkillSense**, a full-stack, AI-driven career intelligence platform designed to bridge the widening gap between candidate capabilities and recruiter expectations in the modern job market. By combining a metadata-aware Retrieval-Augmented Generation engine with multi-stage LLM inference, SkillSense delivers ATS evaluations that achieve a Pearson correlation of $r = 0.90$ with expert human judgements and reduce skill-requirement hallucination by 89.7% compared to retrieval-free baselines. Its ATS-parameterised adaptive mock-testing system improves diagnostic accuracy from 0.65 to 0.85 Pearson correlation and raises candidate-perceived relevance by a mean of 1.4 Likert points. Its browser-native proctoring module reduces dishonest behaviour incidence from 34.4% to 5.3% without requiring any proprietary surveillance software.

Together, these results demonstrate that RAG-grounded evaluation, adaptive assessment, and lightweight browser-based proctoring constitute a practical, scalable, and effective framework for next-generation career intelligence. SkillSense makes a meaningful step toward a future in which every candidate receives honest, specific, and actionable feedback on how to close the gap between where they are today and where they aspire to be.

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